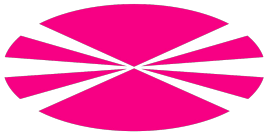




Machine Learning Techniques for Cancer Diagnosis Assistance



UNIVERSIDADE
DA CORUÑA

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Master's thesis 2023/2024

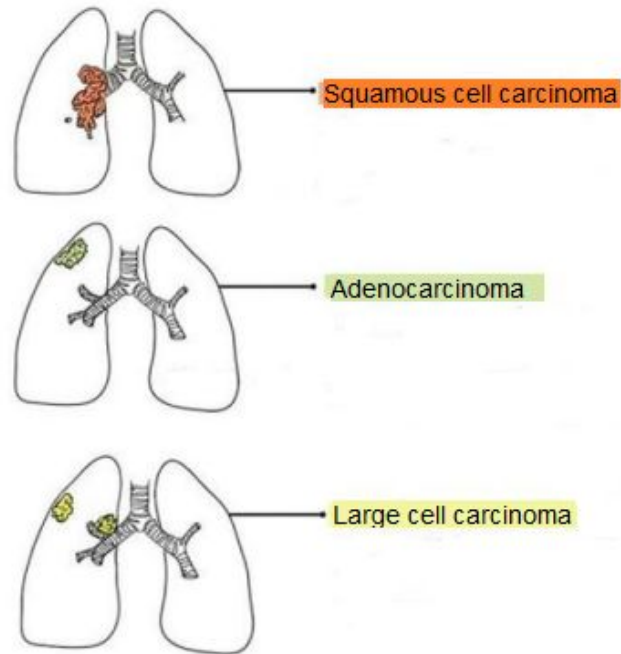
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Context

Non-Small Cell Lung Cancer (NSCLC) [1]

- It represents 13% of all new cancer diagnoses.
- It is one of the diseases with the highest mortality rate.
- Difficulty in early diagnosis.



Novel non-invasive technique

Tumor-Educated Platelets (TEPs) [2]

- They actively interact with the tumour.
- It produces measurable molecular and functional changes.
- Easily detectable by liquid biopsy.

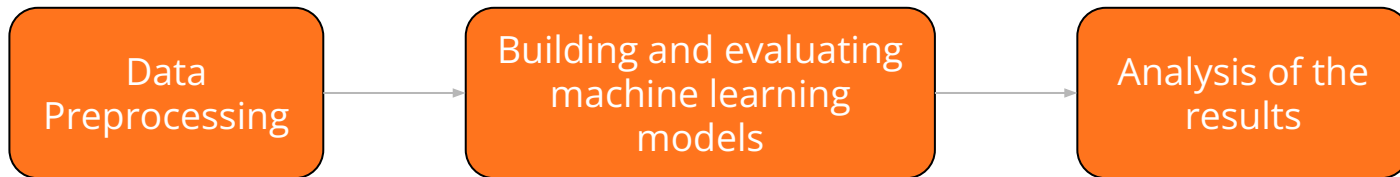


RNA-Seq

	Sample_1	Sample_2	...	Sample_m
gen_1	y_{11}	y_{12}	...	y_{1m}
gen_2	y_{21}	y_{22}	...	y_{2m}
...	...	...	...	...
gen_g	y_{g1}	y_{g2}	...	y_{gm}

Context	OBJECTIVES	Dataset	Data Preprocessing	Building and evaluating machine learning models	Results	Conclusions
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Workflow and objectives



- Find outliers and select relevant features for NSCLC detection.
- Implement the data in 11 machine learning models to evaluate their robustness.
- Obtain a robust model that facilitates interpretation by oncologists to assist in clinical diagnosis.
- Demonstrate the effectiveness of this technique for the early diagnosis of NSCLC.

Context	Objectives	DATASET	Data Preprocessing	Building and evaluating machine learning models	Results	Conclusions
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Dataset → NCBI (GSE89843) [3]

RNA-Seq Matrix

- **779 Subjects**
 - 402 with NSCLC
 - 377 without NSCLC
- RNA-Seq samples of 4722 genes

Metadata Matrix

- Age
- Gender
- NSCLC positive / negative
- Hospital of blood sample collection
- Storage time prior to analysis
- Others of lesser interest...

Context	Objectives	Dataset	DATA PREPROCESSING	Building and evaluating machine learning models	Results	Conclusions
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Detection of Outliers

Data Standardisation

Detection of outliers

Feature Selection

Chebyshev → 67 Samples

One-Class SVM → 77 Samples

Local Outlier Factor → 78 Samples

Isolation-Based → 78 Samples

There is no significant improvement by eliminating outliers.

Context	Objectives	Dataset	DATA PREPROCESSING	Building and evaluating machine learning models	Results	Conclusions
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Detection of Outliers

TABLE I.

TEST SET PERFORMANCE AND HYPOTHESIS TESTING PROCEDURES ACROSS DATASETS AFTER APPLYING OUTLIER DETECTION TECHNIQUES.

SVM	Number of outliers found	F1 Score	Shapiro $\alpha = 0.05$	Levene $\alpha = 0.05$	ANOVA $\alpha = 0.05$	Tukey $\alpha = 0.05$
Original Dataset	-	0.836 ± 0.044	$H_0 \rightarrow \checkmark$	$H_0 \rightarrow \checkmark$	$H_0 \rightarrow \checkmark$	$H_0 \rightarrow \checkmark$
Chebyshev	67	0.849 ± 0.038				
One-Class SVM	77	<u>0.857 ± 0.052</u>				
Local Outlier Factor	78	0.849 ± 0.024				
Isolation Forest	78	0.847 ± 0.042				

Context	Objectives	Dataset	DATA PREPROCESSING	Building and evaluating machine learning models	Results	Conclusions
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Feature Selection

Data Standardisation

Detection of
Outliers

**Feature
Selection**

We removed
low-variance genes
that introduce noise
in the dataset.

Original → 4722 Genes

ANOVA F → 195 Genes

Random Forest → 300 Genes

mVRS → 465 Genes

Double RBF → 330 Genes

**We selected the set of features
identified by Random Forest.**

**This subset accounts for 6.35%
of the total feature set.**

Context	Objectives	Dataset	DATA PREPROCESSING	Building and evaluating machine learning models	Results	Conclusions
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Feature Selection

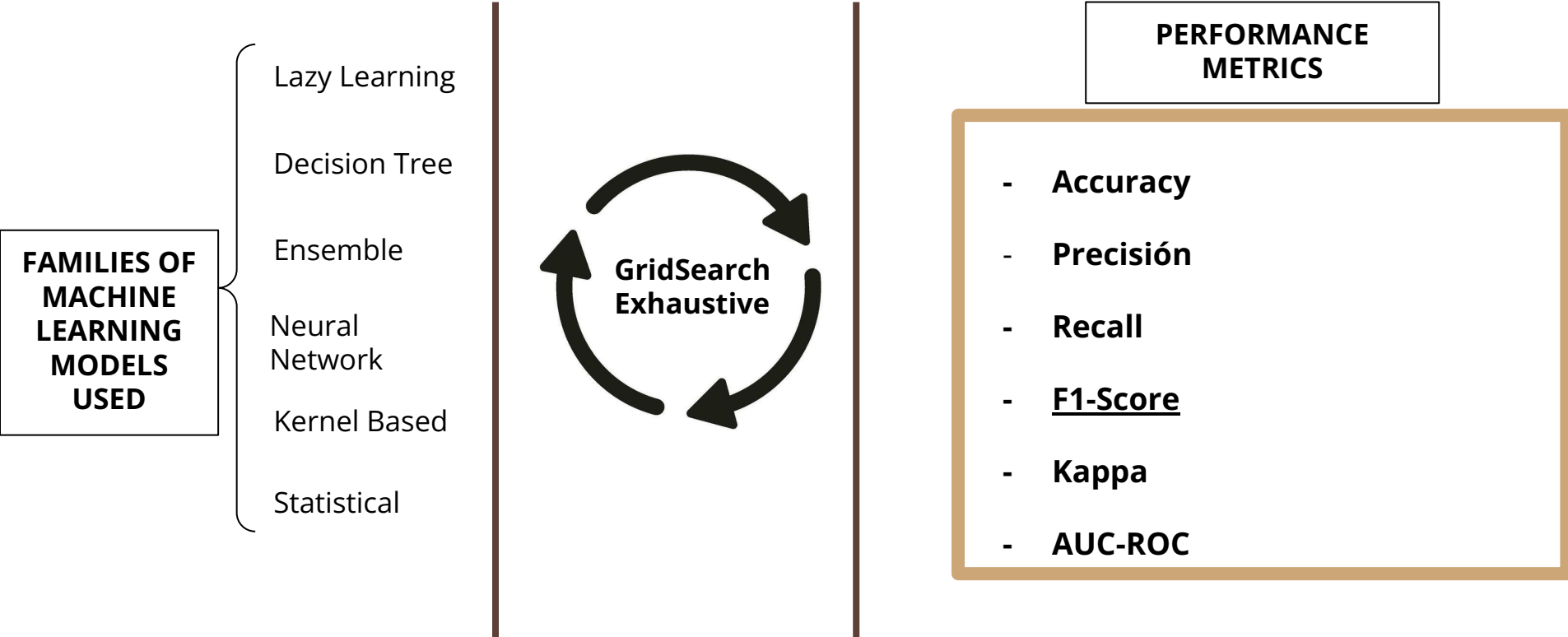
TABLE II.

TEST SET PERFORMANCE AND HYPOTHESIS TESTING FOR THE DIFFERENT DATASETS AFTER APPLYING FEATURE SELECTION TECHNIQUES.

SVM	Number of features selected	F1 Score	Shapiro $\alpha = 0.05$	Levene $\alpha = 0.05$	ANOVA $\alpha = 0.05$	Tukey $\alpha = 0.05$
Original Dataset	4722	0.836 ± 0.044	$H_0 \rightarrow \checkmark$	$H_0 \rightarrow \checkmark$	$H_0 \rightarrow \times$ $H_a \rightarrow \checkmark$ $p\text{-value} = 0.026 < \alpha$	$H_0 \rightarrow \times$ $H_a \rightarrow \checkmark$ $p\text{-Adj} = 0.0158$ between mVRS & Random Forest
ANOVA F	195	0.850 ± 0.047				
Random Forest	300	<u>0.869 ± 0.049</u>				
mVRS	465	0.804 ± 0.047				
Double RBF	330	0.821 ± 0.044				

Context	Objectives	Dataset	DATA PREPROCESSING	BUILDING AND EVALUATING ML MODELS	Results	Conclusions
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Building and evaluating machine learning models



Context	Objectives	Dataset	Data Preprocessing	Building and evaluating machine learning models	RESULTS	Conclusions
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Results of the experimental study

Cross-Validation

kFold = 10

TABLE III.

TEST SET PERFORMANCE FOR THE DIFFERENT MACHINE LEARNING MODELS.

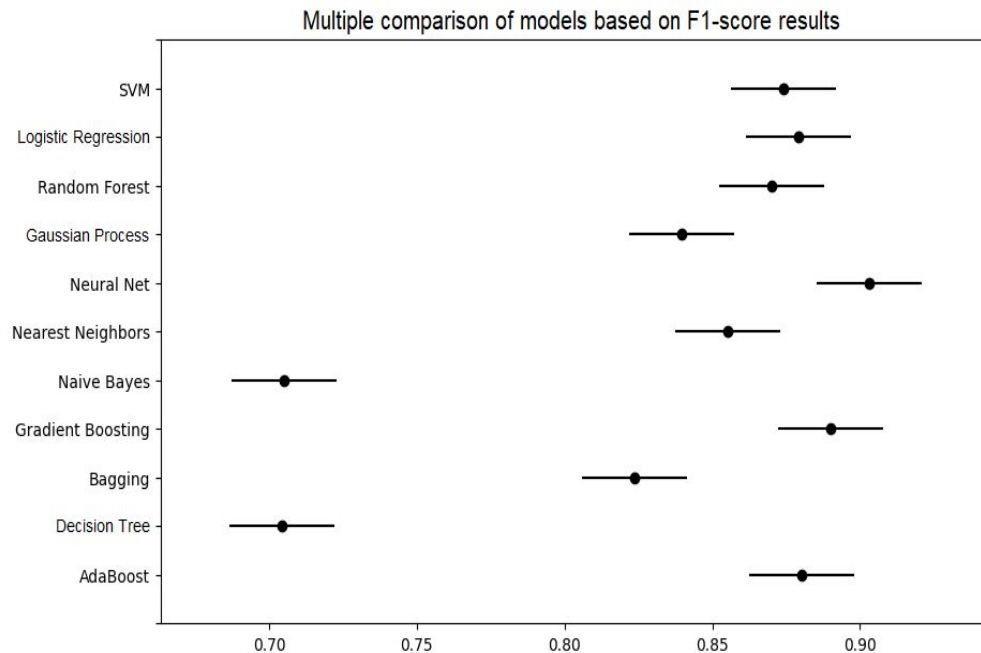
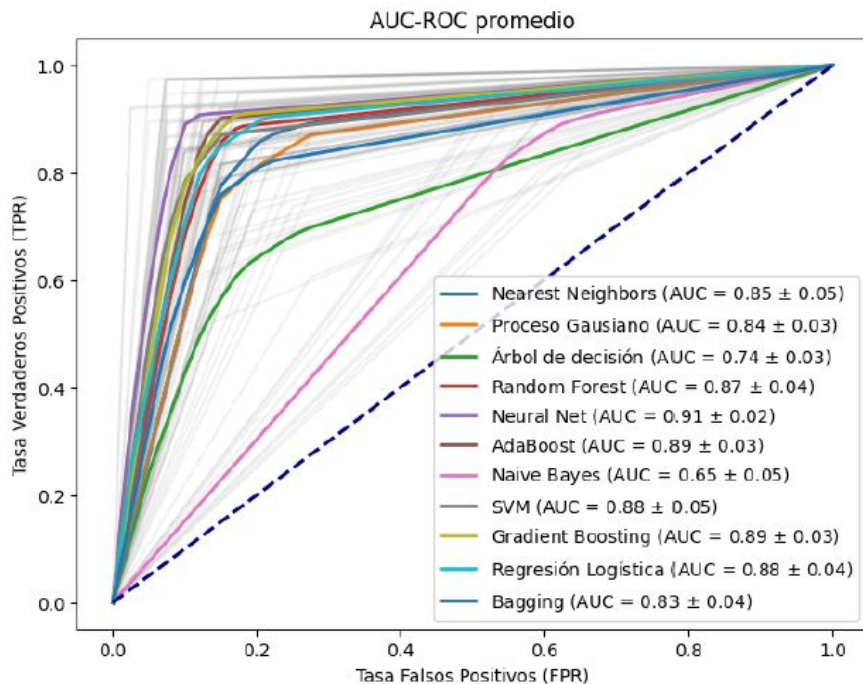
	Accuracy	Precision	Recall	F1 Score	Kappa	AUC ROC
AdaBoost	0.879 ± 0.026	0.867 ± 0.036	0.883 ± 0.038	0.874 ± 0.030	0.785 ± 0.052	0.890 ± 0.026
Decision Tree	0.730 ± 0.036	0.739 ± 0.049	0.682 ± 0.072	0.706 ± 0.045	0.457 ± 0.074	0.739 ± 0.032
Bagging	0.840 ± 0.047	0.849 ± 0.031	0.810 ± 0.056	0.829 ± 0.034	0.678 ± 0.056	0.834 ± 0.038
Gaussian Process	0.840 ± 0.036	0.819 ± 0.039	0.857 ± 0.035	0.836 ± 0.027	0.628 ± 0.072	0.840 ± 0.030
Gradient Boosting	0.891 ± 0.040	0.879 ± 0.040	0.897 ± 0.065	0.887 ± 0.044	0.782 ± 0.052	0.892 ± 0.034
<u>Logistic Regression</u>	0.882 ± 0.021	0.855 ± 0.025	0.908 ± 0.021	<u>0.880 ± 0.022</u>	0.764 ± 0.041	0.884 ± 0.040
Naive Bayes	0.653 ± 0.041	0.590 ± 0.045	0.905 ± 0.038	0.713 ± 0.047	0.319 ± 0.081	0.653 ± 0.048
Nearest Neighbors	0.730 ± 0.036	0.853 ± 0.049	0.822 ± 0.072	0.851 ± 0.045	0.706 ± 0.074	0.854 ± 0.045
<u>Neural Net</u>	0.913 ± 0.019	0.906 ± 0.026	0.915 ± 0.023	<u>0.909 ± 0.021</u>	0.826 ± 0.032	0.907 ± 0.017
Random Forest	0.873 ± 0.025	0.861 ± 0.024	0.879 ± 0.028	0.869 ± 0.024	0.746 ± 0.047	0.874 ± 0.036
SVM	0.881 ± 0.023	0.888 ± 0.026	0.870 ± 0.024	0.875 ± 0.021	0.761 ± 0.048	0.881 ± 0.045

Context	Objectives	Dataset	Data Preprocessing	Building and evaluating machine learning models	RESULTS	Conclusions
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Results of the experimental study

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kFold = 10



Context	Objectives	Dataset	Data Preprocessing	Building and evaluating machine learning models	Results	CONCLUSIONS
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Conclusions

- We demonstrate the viability of using machine learning techniques for the early detection of NSCLC through RNA-Seq data from TEPs, confirming the potential of this approach in clinical practice and oncology research.
- In addition to its performance, Logistic Regression provides a clear interpretation of model decisions, making it a valuable tool for clinical settings where interpretability is crucial.
- The implementation of models in routine medical practice could speed up the diagnosis of NSCLC.

FUNDAMENTAL BIBLIOGRAPHY

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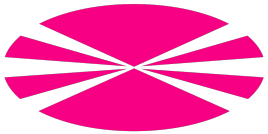
Thank you for your attention!

"I think the biggest innovations of the 21st century will be at the intersection of biology and technology. A new era is beginning"

Steve Jobs



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